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5 April 1985
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, MOBILE BLOOD DONOR, WITH COMPONENT AND PROCESSING LABORATORY, 4-BED, DIESEL ENGINE DRIVEN, 33,000 POUNDS GVW, 4x2, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a completely equipped, self-propelled, self-contained, mobile unit designed to serve as a combination blood component laboratory, 4-bed donor unit, and processing laboratory. The vehicle shall be diesel engine driven, fully air conditioned and heated and mounted on a four wheel, two wheel drive forward control bus type chassis. The vehicle shall have a minimum gross vehicle weight (GVW) of 33,000 pounds. The vehicle shall be furnished with a 45 kilowatt (kW) generator. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

A 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GBS.1, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS
FEDERAL

O-E-915	- Extinguisher, Fire, Dry-Chemical (Hand Portable).
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
* ZZ-T-381	- Tires, pneumatic, Vehicular (Highway).

STANDARDS
FEDERAL

* FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, INC.
SAE Standards and Recommended Practices

	J537	- Storage Batteries.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (20–1000 MHz).
	J588	- Turn Signal Lamps.
	J589	– Turn Signal Switch.
*	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
*	J688	- Truck Ability Prediction Procedure (DoD adopted).
*	J695	- Turning Ability and Off Tracking - Motor Vehicles.
	J816	- Engine Test Code - Spark Ignition and Diesel.
*	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies should be addressed to SAE, INC. , 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color shall be the manufacturer's standard gloss white. The identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

* 3.1.1.2 Special markings. When specified (see 6.2), special markings shall be applied to the vehicle exterior.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit ("F).

* 3.1.1.5 Coolant recovery system and level indicator. A radiator coolant recovery system or surge tank of adequate size to prevent loss of coolant under conditions specified in 3.1.1.4 shall be furnished. A

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translucent reservoir, sight gage or other means for checking the coolant level shall be provided to enable the operator to check the coolant level while standing on the ground. In addition, a low coolant level and a high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

* 3.1.1.6 Drain plugs. Drain plugs installed in manual transmission and rear axle shall be of the permanent magnet type.

* 3.1.1.7 Wood treatment. As specified herein (see 3.5.4), wood shall be treated in accordance with MIL-STD-1223.

3.1.1.8 Towing devices. Not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the rear of the vehicle. Towing devices and rear of vehicle shall withstand, without any signs of failure, lifting the rear end of the vehicle, turning the vehicle within its minimum turning radius, and backing or pulling the vehicle on a level roadway. Decals shall be provided on front and rear bumpers stating "LIFT AND TOW FROM REAR OF VEHICLE."

3.1.1.9 Front tow attachments. Two tow chain attachment devices shall be provided at the front of the vehicle to allow recovery of the vehicle from a ditch or other impediment. Each front attachment device shall provide an ultimate strength at least equal to the GVW rating of the vehicle furnished.

3.1.1.10 Towing brakes. In addition to the brake requirements specified herein (see 3.4.11), an emergency system for controlling at least the front brakes of the vehicle from a towing vehicle shall be provided. The installation shall be complete with air brake couplers; relay emergency valve with no-bleed-back feature, except when spring applied emergency brake is furnished; and additional air lines and fittings. Towing brake installation shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. Couplers shall be mounted in a protected location at the rear of the vehicle and shall permit ready attachment of air hoses from the towing vehicle. Service coupler shall be located on the curbside and emergency coupler on the streetside. Couplers shall be identified.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include weight of the chassis and body with all attachments, accessories and equipment; and full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator and an assistant (weight computed at 175 pounds each), plus a payload of not less than 5,000 pounds, evenly distributed in the body load area. Additionally, the chassis manufacturer's GVW rating shall be not less than 33,000 pounds. The required GVW shall be the greater of the two.

* 3.2.6 Ground clearance. The ground clearance shall be not less than 9-1/2 inches under the axles and not less than 18 inches under all other parts.

3.2.7 Ratings. Vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity.

3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit ready accessibility for routine servicing and shall permit accessibility for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with vehicle loaded with specified payload or to specified GVW, whichever GVW is greater.

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3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades of not less than 1.0 percent at 55 tiles per hour (mph). Gradeability requirements shall be met with the transmission in direct drive and, when a multispeed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed and shall provide a vehicle speed not greater than 4 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded required GVN, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to required GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.3.3 Turning radius. The turning radius, as defined in SAE J695, shall be not greater than 45-1/2 feet.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck, which meets or exceeds the requirements of this specification,

* 3.4.1.1 Diesel engine. The engine shall be a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel type, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.2 Oil filter. A full flow type oil filter shall be furnished.

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longitudinal position on a grade of not less than

3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

3.4.1.4 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. Control shall include partial thermostatic control of coolant flow through the radiator and radiator shutters, or complete thermostatic control of all coolant flow through the radiator.

3.4.1.5 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 120-volt alternating current (at), and shall be wired through a junction block to a single three pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1,500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening.
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.
(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet

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temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt starting *system,* with a 12-volt direct current (de) lighting system, shall be furnished. Unless otherwise specified (see 6.2), a minimum 145-ampere alternator shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, shall have a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

* 3.4.2.2 Lighting. All vehicle (chassis and body) lights, reflectors , and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26. Electrical system shall be 12 volts. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or vehicle bumpers.

* 3.4.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Vehicle shall be provided with single-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588. Operating units shall be provided with a visible and audible flash indicator. Turn signals shall be of the self-cancelling type.

* 3.4.2.4 Batteries. The batteries shall be 12-volt potential. The batteries shall have a total reserve capacity rating of not less than 500 minutes and a total cold cranking rating at O°F of not less than 1800 amperes, both measured in accordance with SAE J537. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. The batteries shall not be exposed to exhaust heat. Batteries shall be mounted, in an enclosed compartment , on a sliding tray with rollers. The sliding tray shall support the batteries in pulled out position. The tray shall be coated with an acid resistant material. A battery access door shall be provided on the side of the body.

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performed with the fuel cubic and hydraulic systems op-

3.4.2.5 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. A single-stage or two-stage dry type air cleaner shall be furnished. The air cleaner shall include an indicator or dash mounted light to indicate when filter should be changed.

* 3.4.3.2 Fuel tanks. Dual safety fuel tanks conforming to Federal Motor Carrier Safety Regulations 393.65 and 393.67 shall be furnished. Each tank shall have a capacity of not less than 65 gallons. Means shall be provided to assure equalized fuel levels in both tanks. The tanks shall be mounted on each side, as far forward as possible. The filler shall be identified and be arranged for safe and convenient filling from the exterior body. The access for the filler neck shall be designed to prevent the entry of spilled fuel to any point inside the vehicle body or entrapment between body walls. Sealing may be accomplished by use of a neoprene rubber thimble or grommet, or other device installed in the access hole.

3.4.4 Exhaust system. The exhaust system, conforming to Federal Motor Carrier Safety Regulation 393.83, shall be substantially constructed, effectively leakproof, and quiet in operation. Properly supported exhaust tailpipe shall extend to the rear bumper and shall be located to properly and effectively minimize the possibility of exhaust fumes entering the body. The exhaust system shall include muffler(s).

3.4.5 Transmission. The transmission input torque capacity shall be at least equal to the maximum torque delivered by the engine.

3.4.5.1 Manual transmission. Unless otherwise specified (see 3.4.5.2), the manufacturer's standard manual synchromesh transmission shall be furnished. The transmission shall provide one reverse speed and not less than five forward speeds. Gear ratios in the transmission and axle shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide the required vehicle performance. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.101 Clutch. The clutch shall be the largest capacity clutch offered for the size of engine furnished with clutch torque capacity exceeding maximum delivered engine torque.

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* 3.4.5.2 Automatic or semiautomatic transmission. When specified (see 6.2), an automatic or semiautomatic transmission shall be provided. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Propeller shaft guards, similar to those required for buses by Federal Motor Carrier Safety Regulation 393.89, shall be provided.

3.4.7 Chassis frame. The chassis frame shall be of the forward control heavy duty type. Chassis frame shall be the manufacturer's standard for the vehicle furnished, and shall be adequate for mounting the body and equipment specified herein. The frame shall be continuous from bumper to bumper. The frame rails shall have a section modulus of not less than 12.98 inches cubed for their entire length. The frame shall be reinforced full length. The resisting bending moment shall be not less than 649,000 inch-pounds. Frame modifications shall be as recommended by the chassis manufacturer in his body builder's book.

3.4.8 Suspension. The vehicle shall be equipped with manufacturer's standard or optional suspension system with components having a rated capacity at least equal to the load imposed on each member , measured at the ground, with vehicle loaded to required GVW. The vehicle shall be provided with spring, full air, or combination spring and air suspension on the front axle. The rear suspension shall be full air. Hydraulic double-acting shock absorbers shall be provided on both the front and the rear axles. Suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the vehicle from the applicable jacking location. Air systems shall incorporate leveling valves with time delay to avoid constant air consumption. Shock absorbers and airbags shall be fabricated of materials capable of operating in ambient temperatures from minus 50°F to plus 125°F.

3.4.8.1 Front suspension. The vehicle shall be equipped with spring, full air or combination spring and air front suspension system. If leaf type springs are used, spring design shall provide a double-wrap spring eye, at least at the stationary end of each spring. Front suspension design shall be such that the natural frequency of vertical

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vibration at the front wheels under all conditions of loading shall not exceed 2.0 hertz. Torque rods shall be provided, if available as manufacturer's standard or option to increase longitudinal and lateral stability.

3.4.8.2 Full air rear suspension. The rear axle with full air suspension shall have four air springs of adequate diameter. The suspension design shall be such that the natural frequency of vertical vibration about any transverse axis shall not exceed 1.5 hertz at rated capacity for required GVW. Full air system shall have none of the spring weight carried on anything other than air springs. No other spring medium such as leaf, coil, or rubber springs may be used in conjunction with the air springs. The full air system shall incorporate four radius rods to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to chassis and body. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication. Suspension system shall include locking devices to lock up the suspension to preclude damage when lifting and towing the vehicle from the rear.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when vehicle is loaded to required GVW. The drive axle shall be furnished at the front or the rear. Torque rods shall be furnished for the rear axle(s) if available as manufacturer's standard or option. Axle gear ratio furnished shall enable vehicle performance required in 3.3.1. Unless a two-speed axle is specified, a single reduction drive axle shall be furnished.

3.4.9.1 Two-speed axle. When specified (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. Gear ratio shall provide performance specified in 3.3.1.

3.4.9.2 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels, Rim size, tire size, type and load range (ply rating) shall be the same for all wheels on the vehicle. Tire and rim ratings shall conform to Tire and Rim Association recommendations, for the type and size of tires furnished. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

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* 3.4.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Steel belted radial ply tires or, when specified (see 6.2), bias ply tires, shall be furnished. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to required GVW. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Balancing. Each tire shall balance within practicable limits. Wheels, hubs, and brake drums shall be effectively in balance. Balancing shall be adequate to preclude wheel shimmy at all vehicle speeds.

* 3.4.10.3 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in an underbody skirt compartment. Threaded fasteners, when used to secure the spare tire in the carrier, shall. be constructed of or plated with corrosion-resistant. material. The carrier design shall enable removal or mounting of spare wheel assembly using only the tools specified in 3.4.16.1.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall furnished. When a spare tire assembly is specified (see 3.4,10.5), a spare wheel. or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare t-ire mounted on the spare wheel or rim. The spare tire shall be of the same size and tread design and of the same load range (ply rating) as furnished on the vehicle.

3.4.10.6 inner tubes. For tube type tires, inner tubes shall be heavy duty type and shall be the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), and 393.44 through 393.52.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader--head type, engine driven and lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm).

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(b) Air storage reservoir(s) with a total capacity of not less than 4400 cubic inches; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
(c) Foot control, suspended or treadle type.
(d) Air control valves.
(e) Air pressure gage, visible to the driver.
(f) Low air-pressure warning, visible and audible.
(g) Service brake stop lamp switch.
(h) Automatic moisture ejector.
(i) Alcohol aspirator with unbreakable transparent container.
(j) Relay emergency valves.
* (k) Automatic slack adjusters.
* (1) When specified (see 6.2), in lieu of subparagraph (i), an air dryer shall be furnished. The air dryer shall be the desiccant type.

3.4.12 Driver's compartment. Driver's compartment shall be built as an integral part of the body and shall include the following:

* (a) Driver's seat shall be a Bostrum, or equal, hydraulic seat, adjustable fore and aft, and for height. The seat shall be positioned for convenience and safe operation of all controls by the driver.
(b) An upholstered, tilt passenger seat shall be provided.
(c) Driver's and passenger's seats shall be equipped with retracting seat belts and fold-down arm rests.
(d) An upholstered seat shall be provided on the top of the engine cover.
* (e) The color of the upholstery and the interior shall be compatible. White upholstery shall not be furnished.
(f) Driver's area floor shall be covered with ribbed rubber matting.
(g) Two storage cabinets shall be installed above the windshield and shall include a center-mounted, battery operated clock. Cabinet doors shall be provided with positive closing latches. The base of the storage cabinets and the top section of the instrument panel shall be padded and upholstered with heavy duty vinyl.
(h) Two 12-volt dome lights shall be installed in the ceiling and shall be provided with individual switch controls.
(i) A hinged door shall be provided on the driver's side, shall include a vertical sliding window and shall be equipped with a lock.
(j) A safety grab rail and recessed steps of safety treadplate shall be provided to facilitate entrance and exit of the driver.
(k) A ventilating window shall be provided on the passenger side.

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3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual, electric windshield wipers with not less than two operating speeds. Wiper system furnished shall sweep the maximum practicable area of the windshield surface. Windshield washers shall be provided. Washers shall be electrically operated with the control on the instrument panel.

3.4.15 Bumpers. A channel type front bumper shall be provided. The rear bumper shall conform to Federal Motor Carrier Safety Regulation 393.86. The top of the front bumper shall. be provided with a skid resistant finish to assist personnel in windshield cleaning (see 3.4.19).

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, electrical cables, portable flood lights with brackets, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools shall be located in an underbody skirt compartment, shall be weatherproof, and shall include a flushmounted lock or provision for padlocking.

3.4.16.1 ToolS. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with a spare assembly, and shall include at least a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The driver’s area shall be provided with a heater of not less than 32,000 British thermal units (Btu) per hour rated capacity. The heater shall be installed with outlets to the floor to provide heat for both the driver and the passenger. A defroster attachment shall be furnished to provide a flow of heated air over the entire windshield area. The heater shall include blower and thermostatic controls mounted convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver when in a seated position. Lever controls shall be designed and so located as to

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permit easy entrance and exit of operator to and from driver's seat. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing, Instruments shall be panel mounted and shall be clearly visible to the driver when seated in normal driving position.

3.4.19 Windshield cleaning grab handles. A handle shall be furnished at each side of the windshield to provide a handhold for windshield cleaning while standing on the bumper (see 3.4.15).

3.4.20 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator, or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Tachometer
(h) Indicator light to show headlight high beam operation
(i) Dual sunvisors
(j) Driver's compartment ventilator other than window
(k) Ash receptacle.

3.4.21 Rearview mirrors. Outside rearview combination (flat and convex) type mirrors shall be furnished. Each combination mirror shall have at least 50 square inches of flat reflective area and a convex surface having at least 20 square inches of reflective area. One mirror shall be mounted on each side of the vehicle. Mirrors shall be mounted to permit maximum vision without obstruction and shall be visible on the right through the passenger side cab window or the upper corner of the windshield and on the left through the driver's window(s). An inside rearview mirror shall be provided.

3.4.22 Horn. An electric horn shall be provided. When specified (see 6.2), in addition, an air operated horn shall be provided.

3.4.23 Automatic speed control. When specified (see 6.2), a control, capable of maintaining the speed of the vehicle at a constant rate set by the operator, shall be furnished. Provisions for engaging and disengaging the control shall be located convenient to the driver. The control shall automatically disengage upon application of the brakes.

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3.5 Body. The body shall be mounted on a forward control bus type chassis with seating for the driver and one passenger forward of the front axle. The unit shall be manufactured in accordance with the highest standards of the industry. The body shall have a generally symmetrical appearance with full-skirted sides and a flat rather than a stepped front design. Body design shall effectively exclude engine fumes and exhaust gases from the interior and shall provide safe and comfortable transportation for the passengers and the driver. All equipment and furnishings shall be secured to withstand mobile vehicle operations.

* 3.5.1 Body dimensions. The vehicle body exterior and interior dimensions shall be as follows:

(a) Overall width, exclusive of mirrors, lights and reflectors, shall be not more than 96 inches.
(b) Overall height shall be not more than 154 inches.
(c) Overall length shall be not more than 40 feet.
(d) Interior length behind driver's seat shall be not less than 34-1/2 feet.
(e) Interior width shall be not less than 90 inches.
(f) Interior height shall be not less than 84 inches.

* 3.5.2 Construction. The body and roof framing shall be constructed of aluminum. extrusions of nominal 0.125 inch thickness or not less than 1-1/2 inch square tubular high tensile steel. The body framing shall be braced at all points of stress. Corner posts shall be aluminum extrusions of nominal 0.1 inch thickness, where aluminum framing is used. Door and window frames shall be reinforced and gusseted to prevent torsional twisting. Body and floor frame members shall be installed on nominal lb-inch centers. The body shall be custom built as specified, providing mobile laboratory facilities to serve as a combination blood component laboratory, 4-bed donor unit and processing laboratory. The vehicle body shall provide a comfortable environment for donors and working personnel for extended periods. The body structure shall be capable of resisting all generally and locally applied loads resulting from normal intended usage without excessive deflection, cracking, permanent deformation, loosening of joints, or premature fatigue failure. Where aluminum and steel surfaces come into contact, effective means shall be provided to preclude electrolytic action.

* 3.5.2.1 Exterior panels. Exterior side, end, and roof panels shall be constructed of aluminum sheets of an alloy and thickness adequate to provide the required strength. The roof panel may be 20 gage (.0359 inch) hot rolled steel panel. The exterior body panels

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shall be all smooth finish or the bottom half of the panels may be beaded. Sheets shall be overlapped and double sealed with Alumelastic sealing compound and shall be securely riveted to the body frame members. The roof shall be a one-piece panel with a rise at the center for efficient drainage. A drip molding shall be provided around the entire perimeter of the body. The body, roof, and panel joints shall be watertight and leakproof. The body exterior side and roof panels and all adjoining surfaces shall prevent the entrance of water or moisture when exposed to a water spray test (see 4.3). Except where foamed-in-place insulation is used, a sound-deadening material shall be applied to the interior surfaces to eliminate drumming. Heavy gage metal tapping plates shall be installed between the frame members to secure wall mounted equipment. Unless toggle bolts are used, plywood tapping plates shall be installed to secure ceiling mounted fluorescent light fixtures. Supports and reinforcements shall be provided for the installation of cabinets, lights and equipment.

* 3.5.2.2 Doors. In addition to the driver's door, two entrance and exit doors shall be provided. The doors shall be of aluminum faced plywood core, one piece molded fiberglass plastic laminate construction, or pan type, constructed of 0.125 inch aluminum, with aluminum tube reinforcement. The doors shall be insulated and shall close on dust-sealing neoprene extrusions. The doors shall include safety glass windows with removable curtains, and double-acting door latches with keyed locks. Doors shall be equipped with chromed grab handles. Door hinges shall be chrome plated, reach-in strap type or stainless steel continuous type. Both doors shall be on the curbside, one located near the front of the vehicle and one located within the reception area (see figure 1).

3.5.2.3 Windshield. A windshield affording maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body. Glass shall be tinted, heat absorbing safety glass.

3.5.2.4 Windows. A total of seven tinted, heat absorbing, sliding safety glass windows with screens shall be installed within the body sides, three on the curbside and four on the streetside in accordance with figure 1. The windows shall be provided with fingertip control permitting easy opening and adjustment to various desired positions. The windows shall be rattleproof and, when closed, rainproof, waterproof and windproof. All windows shall be equipped with curtain rods and curtains of appropriate color to complement the interior decor. Curtain rods shall be readily removable.

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* 3.5.2.5 Insulation. Insulation shall be provided in the exterior doors, walls, floor and ceiling of the body and shall provide resistance to transfer of heat in accordance with the requirements of 3.5.6 and 3.5.7. Insulation shall be either foamed-in-place, flexible or batt type, fibrous or cellular material, having a moisture absorption capacity of less than 2 percent by weight. If foamed-in-place urethane insulation is installed, it shall be of a non-formaldehyde based compound. Materials shall be odorless and fire retardant. Insulation shall not breed nor promote the growth of fungus and shall not cause nor accelerate corrosion of steel, copper or aluminum. Insulation shall be secured to inner side of external walls and provided with means to retain insulation under all operating conditions. Cement, where used, shall be waterproof. The installation of the insulation shall provide. shear strength sufficient to prevent sag of the insulation when assembled vehicle is operated at speeds up to 55 mph over improved roads. A vapor barrier shall be installed between the interior panels and the insulation. A vapor barrier is not required with foamed-in-place insulation. All insulation, to maximum practicable extent , shall be installed prior to the closing of spaces in body sides, front and rear panels, ends, doors, upright body members, roof, roof bows , and other parts.

* 3.5.3 Interior panels. The interior of the body sides, front, rear and ceiling shall be insulated (see 3.5.2.5) and paneled. Foam tape (or equivalent) shall be installed on frame members, prior to the installation of the interior paneling. The interior walls shall be lined full width, height and length with vinyl-coated plywood paneling. Pressed wood shall not be furnished. Panels shall be installed and secured with joiners, edgings and moldings. The ceiling shall be surfaced with Armstrong Reflex V, or equal, hardboard panels.

* 3.5.4 Floor. Floor crossmembers shall be of adequate size to support body and payload without fracture or failure under all combinations of expected loading and shall conform to Federal Motor Carrier Safety Regulation 393.84. Additional crossmembers shall be provided, if required to support floor loads in critical stress areas. Galvanized sheet shall be installed to fully enclose the floor area. Insulating hardboard shall be secured above the galvanized sheet. Waterproof plywood shall be installed above the insulating hardboard. The unit shall be constructed without wheelhousing to provide a straight-through floor. Underside of floor panels, exposed to road, shall be sprayed with sound deadening, vibration dampening material. Wood floor parts, including plywood, shall be treated as specified in 3.1.1.6.

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* 3.5.4.1 Entrance/exit floor area. A straight floor shall be furnished at the entrance/exit door in the reception area (see figure 1). The entrance/exit door shall be provided with a four step, pull out step assembly that is flush with the entrance/exit floor when in use, The step assembly shall slide under the floor with provisions for locking securely in the travel position. An inset stepwell with pull-out steps shall be provided at the entrance/exit door in the donor area (see figure 1), with provisions for locking securely in the travel position. All stair treads shall be surfaced with non-skid materials. Removable safety handrails shall be provided.

* 3.5.4.2 Floor covering. The entire floor except for the driver's area and the donor area shall be surfaced with Armstrong, or equal, 100 percent nylon, industrial grade carpet with foam back. The floor in the donor area shall be surfaced with heavy duty, cushioned vinyl linoleum. The coverings shall be properly installed and sealed at the joints.

3.5.5 Electrical system. The electrical system shall be in accordance with the National Electrical Code, (NEC) NFPA No. 70. The unit shall be wired to receive 4 wire, 200 amp, 120/240 volt, 60 Hertz (Hz), ac, single phase power from local power sources via a lead-in cable or from the generator (see 3.5.5.3). Wiring shall be provided for operation of all 120/240 volt equipment. Circuit breaker boxes of adequate capacity shall be installed at appropriate locations within the vehicle.

3.5.5.1 Wiring. All 120/240 volt wiring shall be NEC standard plastic-jacketed , non-metallic sheathed cable with ground. Wiring shall be concealed yet easily accessible for repairs. Wiring and related devices shall be installed in a workmanlike manner, mechanically and electrically secure. Devices, lamps, and other equipment requiring periodic service shall be readily accessible and serviceable. Wiring shall be protected by grommets when passing through any hole or body framework. Wiring shall include the following:

(a) Individual 120 volt circuits for utility equipment, lights, receptacles , and air conditioners.
(b) 240 volt circuits for electrical heating units.
(c) Duplex 120 volt receptacles located as shown in figure 1.
(d) Individual 240 volt receptacles located as shown in figure 1.
(e) One 50 foot, 4 wire lead-in cable, sized to unit capacity, with waterproof receptacle on one end and insulated alligator clips on the other end for connection to a local 120/240 volt power source.
(f) Two 100 foot, 120 Volt$_s$ 20 amp lead-in cables connected to circuits within the mobile unit for overnight operation of refrigerator(s).

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* 3.5.5.2 Lighting. Fluorescent lights in standard 1, 2, or 4 tube fixtures shall be securely surface mounted on interior body ceiling within plastic diffuser panels or beneath cabinets above work counters to provide a minimum light density of 75 foot candles at working heights in all parts of each area. Light fixtures shall be installed so that working personnel will not cast a shadow on their work. Diffusers shall be equipped with latching devices to prevent falling in transit. Twelve-volt emergency dome lights shall be provided at each entrance/exit door and within each exterior body compartment. The lights shall be mounted by use of toggle bolts or plywood backing plates not less than 3/4 inch thick.

* 3.5.5.3 Generator. A 45 kW, 4-wire, 120/240-volt, 60 Hz, ac, single phase generator shall be provided. The generator shall be shock mounted on vibration isolators and shall be installed for easy servicing.

* 3.5.5.3.1 Generator engine. The generator shall be driven by a liquid cooled diesel engine. The generator engine shall have sufficient output power to drive the generator to meet the requirements of 3.5.5.3. Fuel for the generator engine shall be supplied from the vehicle fuel tanks. The engine shall. be equipped with starting battery, fuel pump, air cleaner, and critical area silencer. The generator engine shall be installed for proper heat and exhaust dissipation. The engine mounting shall minimize transfer of vibration. A seamless exhaust section shall connect the generator engine manifold with the intake of the critical area silencer. A flexible, stainless steel exhaust section, not less than five feet in length, shall be provided. The exhaust section shall be equipped with a quick disconnect connector for extension of the underbody exhaust line for additional. fume dissipation.

3.5. 5.3.2 Generator compartment. A generator compartment shall be located at the rear of the vehicle. The ceiling and the back wall of the compartment shall be surfaced with fireproof panel sheets secured to 0.090 inch thick aluminum panels or 12 gage (0.1046 inch) steel. panel below the floor line and 24 gage (.0239 inch) steel panels above the floor line. Hinged vent panels shall be provided at each side for proper ventilation. The compartment shall be enclosed with not less than l/2-inch thick Homesote, or equal, panels on the body interior and surfaced with Formica, or equal, countertopping and paneling on the interior walls. The area between the generator compartment and the body interior walls shall be insulated. The compartment shall be accessible from the rear of the vehicle via ‘hinged, louvered doors equipped with flushmounted locks and holders for securing doors in open position.

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3.5.5.3.3 Generator accessories. Generator accessories shall include:

(a) A 5-foot pigtail cable section with 200 amp waterproof receptacle for plugging into power inlet receptacle installed within generator compartment.
(b) A remote start/stop button, hour meter, and battery condition indicator installed within body interior.
(c) An ac instrument package.

* 3.5.5.4 Exterior floodlights. Unless otherwise specified (see 6.2), four brackets and four 150 watt portable floodlights shall be provided for night illumination. Waterproof, 120 volt receptacles shall be installed on the curbside body exterior. One shall be located at the front. of the vehicle, one shall be adjacent to the front door, one shall be adjacent to the rear door, and one shall be at the rear of the vehicle. The floodlights shall be removable for storage during transit. The floodlight installation shall be protected by ground fault interrupters.

* 3.5.6 Air conditioning. Four 120 volt roof mounted air conditioners shall be installed. Installation shall include electric resistance type heat strip within reinforced roof section. Each unit shall be rated at 13,500 Btu per hour cooling and 5,100 Btu per hour heating. The air conditioners shall be capable of maintaining not more than 80°F dry bulb and 68°F wet bulb temperature, and an effective temperature of 74°F within each area inside the vehicle when the ambient temperature is 95°F dry bulb. The interior relative humidity shall be maintained at not less than 30 percent and not more than 60 percent. All air shall pass through a filter system. Each unit shall be provided with a flexible tube installed to carry condensate run-off directly to the ground.

* 3.5.7 Heating. Four thermostatically controlled, electric blower heaters shall be installed throughout the body interior. Each heater shall be rated at not less than 6,800 Btu per hour. The four electric blower heaters shall be capable of maintaining the vehicle inside body temperature at a minimum of plus 70°F midway between the floor and ceiling in ambient outside temperature of O°F with no occupants. When specified (see 6.2), two additional electric blower heaters shall be installed, one above each curbside door. Each additional heater shall be rated at not less than 3,400 Btu per hour.

* 3.5.8 Plumbing. A plumbing system shall be installed. The plumbing system shall be designed to receive. water from an installed, 30-gallon fresh water tank. When specified (see 6.2), the plumbing

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system shall also be capable of receiving water directly from local sources. Plumbing fixtures, devices, and appurtenances shall be supplied with water in sufficient volume and at pressures adequate to enable them to function satisfactorily and without undue noise, leakage,

	Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 KHz).
J560	Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	Rear Wheel Splash and Stone Throw Protection.
3697	Safety Chain of Full Trailers or Converter Dollies.
J702	Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer (DOD Adopted).
J706	Rating of Winches.
J844	Nonmetallic Air Brake System Tubing (DOD Adopted).
J931	Hydraulic Power Circuit Filtration.
J1067	Seven Conductor Jacketed Cable for Truck and Trailer Connections.
J1292	Automobile, Truck, Truck-tractor, Trailer, and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

Frame modifications shall be as

not less than 649,000 inch-pounds

e shall be equipped with
suspension system with components
ual to the load imposed on each
n vehicle loaded to required GVW.
spring, full air, or combination
ont axle. The rear suspension shall
ng shock absorbers shall be provided

3.4.8 Suspension. The vehicle
manufacturer's standard or optional
having a rated capacity at least eq
member, measured at the ground, wit
The vehicle shall be provided with
spring and air suspension on the fr
be full air. Hydraulic double-acti

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Velcro or equal holding devices shall be installed. Counters shall be of adequate size for use as donor beds and shall be 30 inches high. Two counters shall be on the streetside and two on the curbside (see figure 1).

(b) Storage cabinets and drawers shall be installed under each of the four work counters (see figure 2). The two counters on the streetside shall have style 1 storage arrangement and the two on the curbside shall have style 2. Positive latches shall be provided for security in transit.

(c) Removable, 10-inch high separators shall be provided between donor beds.

(d) Stainless steel sink shall be install-cd within streetside counter between donor beds.

(e) Four castered upholstered stools shall be provided and shall be secured under the work counters during transit.

(f) A floor-to-ceiling partition with full length curtain shall be provided to separate the donor area from the reception area.

3.5.9.3 Reception area. The reception area shall include the following:

(a) An upholstered bench with backrest and storage under lift-up seat shall be installed on curbside.

(b) Formica, or equal, surfaced desk with center pencil drawer and two storage drawers shall be installed against the partition separating the donor area from the reception area.

(c) Castered upholstered secretarial stool with backrest shall be provided and secured under desk during transit.

(d) A 24-inch by 36-inch one-way glass shall be installed in partition above desk to provide a view of the donor area.

(e) A chair shall be provided for donor registration.

(f) A shelf, 6 inches in depth with a l/2-inch lip on the open sides and a skid resistant surface, shall be mounted on the streetside wall, 10 inches above the top of the desk. The shelf shall extend from the partition rearward to the front edge of the desk.

(g) Coat hooks shall be installed at convenient locations.

(h) A floor-to-ceiling partition with access through a wall mounted sliding door shall be provided to separate the reception area from the laboratory area. The partition shall be 4 inches thick and shall be insulated to increase sound-deadening qualities.

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3.5.9.4 Laboratory area. The laboratory area shall include the following:

(a) A Formica, or equal, surfaced work counter shall be installed above the generator compartment at the rear of the vehicle.
(b) One blood plasma freezer, Puffer-Hubbard Model BVF-7 or equal, shall be provided and secured to the rear work counter.
(c) One blood bank refrigerator, Puffer-Hubbard Model BB-7 or equal, shall be provided and secured to the rear work counter.
(d) Hinged Formica, or equal, surfaced work counters shall be installed adjacent to the rear work counters on the streetside and the curbside. The streetside counter shall be securely mounted to the wall, shall hinge down from the wall to a working level 36 inches from the floor, and shall measure 36 inches in length and 30 inches in depth. The curbside counter shall be securely mounted to the rear access door (see (h)), shall hinge down to a working level 36 inches from the floor, and shall measure 30 inches in length and 30 inches in depth. The counters shall be held in up position by positive holding devices and shall be adequately supported when in working position.
(e) One DuPont Sorvall Model RC-3B with rotor, or equal, automatic refrigerated centrifuge shall be installed next to counter on the streetside. The centrifuge shall be provided with one extra set of cups. Centrifuge unit shall include elevated control panel with combined master switch and circuit breaker, speed selector permitting automatic acceleration to all speeds from 150 rpm to 6,000 rpm, gyro-action self-centering direct drive , heavy duty refrigeration system with temperature control range from minus 20 degrees Celsius (“C) to plus 40°C at full speed, adjustable dynamic braking, electric tachometer, pilot lights, temperature control and indicator. Units to operate from 230 volts ac, 60 Hz, single phase, 25 ampere vehicle supply. Units shall be equipped with 30-minute timer.
(f) Walls within room shall be surfaced with nylon, industrial grade carpet with foam back to increase sound-deadening qualities. The carpet furnished shall be the same as specified in 3.5.4.2.
(g) Six 120 volt exhaust fans shall be installed within the walls for heat dissipation. Two fans shall be located within the curbside wall, one behind the centrifuge and one below the hinged work counter on the streetside wall, and one each behind the refrigerator and the freezer within the rear wall.
(h) Double access doors, providing an opening of not less than 60 inches by 60 inches, shall be installed on the curbside for removal of equipment.

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(i) A wall mounted cabinet shall be installed above the access doors for storage.

3.5.9.5 Additional equipment. When specified (see 6.2), the following additional items of equipment, as specified, shall be furnished:

(a) One blood bank refrigerator, Puffer-Hubbard Model BB-7 or equal, installed in the laboratory area on the streetside below the hinged work counter.
(b) One refrigerated centrifuge with rotor and one extra set of cups, DuPont Sorvall Model RC-3B or equal, installed in the laboratory area on curbside adjacent to laboratory/reception area partition.
(c) One blood bank refrigerator, Puffer-Hubbard Model BB-7 or equal, installed in the laboratory area on curbside adjacent to rear work counter.
(d) Formica, or equal, surfaced counter installed above the refrigerator in recovery area for coffee brewer mounting.
(e) Coffee brewer, including three decanters and three warming plates, secured to the counter above the refrigerator in the recovery area.
(f) Two cell-washing systems with distribution assembly, DuPont "MOHN" or equal, stowed securely within vehicle.
(g) One optical projection balance, Torbal Model PL-12D or equal, stowed securely within vehicle.
(h) One forced air incubator stowed securely within vehicle.
(i) Six Immufuge centrifuges, Dade or equal, stowed securely within vehicle.
(j) Four blood shaker and weight monitor devices with mobile case and calibration weight, SEBRA Model 102 or equal, stowed securely within vehicle.
(k) One Hematron dielectric sealer, Fenwal or equal, stowed securely within vehicle.

3.5.10 Sound system. A sound system including a solid state amplifier, an AM/FM radio/cassette tape recorder unit and four ceiling flush mounted speakers shall be provided. The amplifier shall include a microphone and cord, and connectors to connect additional sound equipment. One speaker shall be installed in each of the four areas specified (see 3.5.10.1 through 3.5.10.4). Speakers shall be provided with individual volume controls and conveniently located to permit easy access. The sound system shall perform satisfactorily with all equipment in operation.

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3.5.11 Fire extinguishers. A five-pound chemical type fire extinguisher conforming to Federal Motor Carrier Safety Regulation 393.95(a) shall be mounted in a convenient location within the reception area near the curbside door. The fire extinguisher shall have an Underwriter's Laboratory rating of 2A:10B:C or more, shall include a metal safety pin and wire, and shall conform to type I, class 1, size 5 of O-E-915. Two 2-1/2 pound chemical type fire extinguishers shall be provided, one mounted in the generator compartment and one mounted in the driver's compartment. Each of these two fire extinguishers shall have an Underwriter's Laboratory rating of 5B:C or more, shall include a metal safety pin and wire and shall conform to type I, class 2, size 2 1/2 of O-E-915. All fire extinguishers shall permit visual determination of condition of charge.

3.5.12 Emergency reflective triangles. Three emergency reflective triangles , conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i), shall be furnished and properly stowed.

3.5.13 Stabilizing jacks. Four electro-hydraulic stabilizing jacks with 12 inch pads shall be provided, one installed under each corner of the body. Each jack shall be rated at 25,000 pounds load capacity and shall be controlled from a central control panel in the body skirt. The jacks shall be capable of leveling and stabilizing the vehicle during stationary service.

3.5.13.1 Sight levels. Sight levels shall be provided at body rear and curbside to aid in leveling unit.

3.5.14 Underbody storage compartments. In addition to the spare wheel carrier compartment (see 3.4.10.3), the tool stowage compartment (see 3.4.16), and the water tanks compartment (see 3.5.9), all remaining available skirt space from the rear of the front wheels to the rear of the vehicle shall be used for additional storage compartments. Compartments shall be waterproof and shall be accessible through a hinged door with holding device and flushmounted chrome lock or provision for padlocking. All compartment door locks furnished shall be operable with the same key.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustments of chassis engine, heating and air conditioning systems, and brake system; filling and charging of all batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engines, and running gear with grades of lubricants recommended for the ambient air temperature at

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the delivery point; servicing of engine cooling systems with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

* 3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All parts shall exhibit no evidence of damage, mutilation, or poor workmanship. Damaged components or parts that have been subjected to rebuild or repairs shall not be used as replacement parts or components. All mounting devices such as bolts, nuts and rivets shall be complete and secured in accordance with the highest standards of industry.

* 3.7.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not degrade the design strength of metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

* 3.7.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

* 3.7.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of an approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and shall be in full contact with the surface of the member.

* 3.7.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

* 3.7.5 Paneling installation. All interior paneling shall be installed uniformly, free of misalinement, open seams, gaps, and uneven surfaces. Interior paneling shall be properly secured, free of misalined joiners, edgings, waviness, dents, nicks, scratches, sharp

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edges, and other forms of deformation or surface defects. Exterior paneling shall be free of damaged or missing rivets. Exterior over-lapped sheet surfaces shall be sealed with sealing compound with uniform application.

3.7.6 Door and window installation. All exterior doors and windows shall be properly located, fitted and sealed to prevent entrance of air, dust and water. All windows and doors with associated hardware shall operate without binding. Window curtains shall be the appropriate color, durable material, free of uneven stitching, crooked seams, runs, tears, or color fade and shall be installed uniformly.

3.7.7 Installation of insulation. The insulation shall be installed in a manner excluding excessive voids, open spaces, rips, tears and other types of damage or defects. Wherever practical, the insulation shall be installed prior to the closing of spaces of the body interior. The insulation shall be installed and secured in a manner to retain the material in a permanent position under all. operating conditions.

3.7.8 Floor installation and floor covering. Galvanized sheet. metal used to enclose the floor area shall be securely installed and free of gaps, open seams, cuts, or open holes. All wood parts shall be free of cracks or splinters and thoroughly treated to provide full and uniform coverage (see 3.504). The material shall be applied to provide uniform coverage, free of thin or bare areas. The carpeting shall be installed securely and fitted without gaps, visible seams, bulges, frayed edges or color fade.

3.7.9 Electrical system installation. The electrical system shall be installd to provide proper routing for electrical wiring with clearances between wiring and applicable parts. All wiring shall be secured and fastened at all points of bending and protected against chafing and damage. Wiring shall be concealed and easily accessible for repairs. All circuits shall be properly grounded. Electrical wiring covering shall be free of cracks, cuts, kinks, frays, or similar defects. All wiring and outlets shall be securely installed, free of loose or faulty connections. Wall outlets and other receptacles shall be conveniently located and installed uniformly. All electrical appliances and fixtures shall be mounted securely and properly positioned to prevent damage during transit.

3.7.10 Air conditioning and heating installation. Air conditioning units shall be securely roof mounted. The units shall be installed, fitted and sealed on the roof surface and shall be rainproof and waterproof. The air conditioning units shall perform satisfactorily

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without vibration or undue noise. The four thermostatically controlled electric blower heaters shall be mounted securely in the specified locations. The heater installations shall be free from gaps, exposed electrical wiring, or insulation. The fan blowers shall operate free from rattles, vibration and any other undue noises or malfunction.

* 3.7.11 Plumbing installation. The installed plumbing system (appliances, fixtures, devices and associated equipment) shall be capable of functioning satisfactorily without water leaks, excessive vibration, or undue noise. All plumbing equipment shall be securely mounted and shall show no evidence of damage (scratches, dents, chipping). All drainage plumbing shall be installed in a manner to preclude leakage.

* 3.7.12 Laminated surfaces. All laminated surfaces (formica covered work counters, desks, etc.) shall be securely installed, properly fitted, alined and leveled with the floor surface. Laminated surfaces shall be free of delamination, scratches, mars, uneven surfaces , rough or chipped edges or corners and color fade. Counters, cabinets and desks shall be of rugged construction, properly alined and installed without any evidence of damage or deformation. Desk and cabinet drawers shall be fitted to provide ease of operation without binding, interference or excessive side movement during opening and closing.

* 3.7.13 Benches, castered stools and chair. The padded and upholstered benches with backrest and fold-down arms shall be of sturdy construction and securely installed. The benches and storage compartment surfaces shall be fitted and alined to provide blended surfaces. The lift-up seat and fold-down arms shall be capable of operation without binding or interference. The castered stools and chair(s) shall be of sturdy construction and shall provide an even contact with the floor.

* 3.7.14 Upholstery and padding. Removable fitted pad coverings and upholstery shall be of durable materials and properly fitted. Upholstery shall meet the flammability requirements of FMVSS 302. All materials shall be free of rips, tears, splits, color fade, or stains. The color of the materials shall be coordinated with the applicable area. The padding material shall be evenly distributed, breathable and free of bumps, bulges, or other uneven surfaces,

* 3.7.15 Grab handles and removable safety handrails. The grab handles and handrails shall be of rigid construction and securely mounted. The surfaces shall be smooth, free of burrs, nicks and scratches.

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* 3.7.16 Sound system installation. The sound system's ceiling mounted speakers shall be securely mounted and fitted, free of gaps, openings , exposed electrical wiring or exposed insulation.

* 3.7.17 Exhaust fan installation. Exhaust fans shall be securely mounted and fitted, free of gaps, openings, exposed electrical wiring or exposed insulation. The installation shall be rainproof and waterproof. The exterior sealant shall be uniform, neat in appearance and free of bare or thin areas. The exhaust fans shall operate without rattles, undue noises or malfunctions.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.3 In-process water spray test. To determine conformance to 3.5.2.1, 3.5.2.3 and 3.5.15, each vehicle produced under the contract, shall be subjected to a water spray test. The water spray test shall be performed prior to installation of interior paneling and insulation. The vehicle shall be exposed to a 100 percent water spray for not less than 15 minutes. The spray shall be delivered by nozzles operating at not less than 25 pounds per square inch (psi) water pressure. The nozzles shall be sufficient in number and placed approximately 4 feet from the body to afford full coverage of sides, roof, front, rear and undercarriage of the body. The spray shall be applied simultaneously or section by section. All doors and windows shall be closed as in normal travel position. During the water spray test, all windows and paneling

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shall be observed for leakage into the interior of the vehicle. The interior of the underbody storage compartments shall be examined for leakage. Any evidence of leakage (water or moisture) shall be cause for rejection.

* 4.4 First production vehicle inspection. To determine conformance to all the requirements of the contract, the first vehicle produced under the contract shall be subjected to in-process water spray test, road test, functional tests, and inspections contained in table I. The inspections and road test shall be conducted by the contractor, at or near his plant, under the direction and in the presence of Government representatives. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the contract.

* 4.4.1 Vehicle weight. To determine conformance to 3.2.4 and 3.2.5, the first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles will be computed using the curb weight, the operator weight at 175 pounds, and the payload to provide the specified GVW. Calculated imposed loads on front and rear axles shall be utilized to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contractual requirements.

* 4.4.3 Road test. To determine conformance to 3.3 through 3.3.3, the first production vehicle with rated payload shall be road tested prior to the water spray test and after completion of the vehicle (see 4.3). The vehicle shall be road tested for a distance of not less than 50 miles. The vehicle shall attain speeds up to at least 55 mph. At least 20 percent of the road test distance shall be over dry, unimproved, or gravel roads at speeds up to 20 mph. The vehicle shall be subjected to not less than three sudden stops from speeds of not less than 20 mph on a substantially level, dry, smooth surface free of loose material. The vehicle shall be observed for the specified stopping distance. The vehicle shall be subjected to a turning radius test. The turning radius shall be not greater than 45-1/2 feet. Upon completion of the road test, the vehicle components and accessories shall be examined for damage and lubricant or coolant leakage. All equipment permanently installed or secured for transit shall be examined for damage.

* 4.4.4 Truck body, treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

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* 4.4.5 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 4.4.6 Functional tests. After completion of the road test, the following functional tests shall be conducted.

* 4.4.6.1 Electrical system test. To determine conformance to 3.5.5 through 3,5.5.3, the electrical system shall be subjected to a functional test. The satisfactory performance of the electrical system shall be verified, with the air conditioners operating, and then all lighting and electrical equipment shall be operated. The lighting shall be observed during the phase-in operation of the air conditioners, to verify that the lighting does not fade and the air conditioning units perform satisfactorily. The power and electric circuits shall be verified to assure that the power load is in accordance with the specified requirements and actual loads imposed.

* 4.4.6.2 Air conditioning test. To determine conformance to 3.5.6, he air conditioning system shall be subjected to a functional test. The air conditioning/vehicle body combination shall be tested under actual or simulated conditions at least as stringent as the specified requirements. Simulated conditions acceptable to the Government may be used. Performance of the air conditioning system shall be verified to determine that the system is capable of maintaining the inside of the vehicle body to an evenly circulated cooling temperature of 72°F maximum in 95°F outside ambient temperature. Inside temperature may be calculated either with the number of occupants required or with the use of simulated heat for sensible cooling.

* 4.4.6.3 Heating test. To determine conformance to 3.5.7, the vehicle body heating system shall be subjected to a functional test. The electric fan blower heaters and vehicle body shall be tested under actual or simulated conditions at least as stringent as the specified requirements. Simulated conditions must be acceptable to the Government. Performance of the heating system shall be verified to determine that the heating system is capable of maintaining the inside body temperature at a minimum of 70°F midway between the floor and ceiling in O°F outside ambient with no occupants.

* 4.4.6.4 Air conditioning test data. Results of the air conditioning and heating tests performed in accordance with 4.4.6.2 and 4.4.6.3 shall be recorded. Test data including a description of all applicable pertinent test conditions shall be recorded and furnished to the Government representative. If the vehicle manufacturer has

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previously tested an identical model vehicle with the same components within a 12-month period, a certification and the specified test data verifying conformance to this specification may be provided in lieu of the first production vehicle air conditioning or heating test. Certification and previous test results, if furnished, shall be forwarded to the engineering office of the procuring activity with the Commercial Vehicle Engineering Data (questionnaire).

* 4.4.6.5 Water system test. To determine conformance to 3.5.8 and 3.7.11, the water system shall be subjected to a hydrostatic test. The in-place water supply system and water heater with connections shall withstand a pump pressure of not less than 40 pounds per square inch (psi) and not more than 50 psi for a period of not less than 30 minutes. The hydrostatic test shall be conducted with no evidence of leakage and no additional air or water needed to maintain the pressure. During the test period the water system shall be observed for leakage, undue noise, and damage as a result of the test.

* 4.4.6.6 Water drain system test. To determine conformance to 3.5.8 and 3.7.11, the water-drainage system shall be tested for leakage. With the main drain outlet capped, the sink shall be filled with water. After a period of not less than 30 minutes, the drain system shall be examined for leakage. Evidence of any leakage shall be cause for rejection.

* 4.4.6.7 Refrigerated centrifuge test. To determine conformance to 3.5.9.4(e), the automatic refrigerated centrifuge shall be subjected to a functional test. The centrifuge unit with elevated control panel shall be operated to verify that the speed selector permits automatic acceleration to 6,000 rpm. The heavy duty refrigeration system shall be observed to verify the entire temperature control range of O degrees Celsius (“C) to plus 30°C at full speed and temperature read-out from minus 20°C to plus 40°C. The timer shall be observed for satisfactory performance for a period of 30 minutes. Upon completion of the test, the centrifuge unit, all associated equipment, components and accessories shall be examined for defects or damage caused by the test.

* 4.4.6.8 Stabilizing jacks test. To determine conformance to 3.5.13 and 3.5.13.1, the stabilizing jacks shall be tested. With jacks in place, the adjustable jacks shall be operated to level and stabilize the vehicle. The jacks shall be observed for proper operation without binding or malfunctioning. The vehicle shall be observed to verify the jacks will provide adequate leveling and stability in relation to the ground, under normal stationary service.

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4.4.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

* 4.4.8 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for rejection of the vehicle. Further, the Government shall continue to refuse acceptance of all vehicles under the contract, until the contractor has taken corrective action to eliminate the cause(s) of the failure.

* 4.5 Production vehicle inspection. Each production vehicle shall be subjected to dimensional inspection and operational tests as specified in section 3. The vehicle, associated controls and equipment shall be operated to the extent necessary to verify compliance to contractual requirements. The contractor's inspection system, as a minimum, shall assure that the vehicle and attached equipment conform to the dimensional and operational requirements and the vehicle is capable of meeting the requirements herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government. The records shall indicate that servicing and adjusting required by 3.6 have been accomplished.

* 4.5.1 Vehicle body inspection. Vehicle bodies installed on the chassis shall be inspected for the characteristics and defects contained in table I. Sampling for inspection shall be in accordance with MIL-STD-105. Acceptance and rejection shall be processed in accordance with acceptance and rejection criteria of MIL-STD-105. Vehicle bodies shall be inspected for conformance to Acceptable Quality Level (AQL) on the basis of percent defective as specified in table 1.

TABLE I. Major defects AQL - 25 defects per 100 units.

Defect number	Characteristic	Major defects	Method of inspection
101	Exterior paneling	Unevenness, buckling, warping, missing or damaged rivets, overlapped panel surfaces not sealed, excessive or sloppy application or sealing material, improper gage metal, surface defects (see 3.5.2.1 and 3.7.5).	Visual

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TABLE I. Major defects AQL - 25 defects per 100 units. - Continued

Defect number	Characteristic	Major defects	Method of inspection
102	Doors and door hardware	Improper fit, not alined, warpage, improper application of door seals, improper material, inoperative door hardware, surface defects (see 3.5.2.2 and 3.7.6).	Visual and operational
103	Sliding windows	Sloppy installation of weatherproof seals, windows not easily opened or adjusted to various positions, defective glass (cracks, scratches) (see 3.5.2.4 and 3.7.6).	Visual and operational
104	Interior panels	Unevenness, buckling, gaps, open seams, joiners, edgings, and molding improperly installed, surface defects (see 3.5.3 and 3.7.5).	Visual
105	Floor and floor covering	Gaps, waviness, cracks, bulges, open seams, inadequately secured, improper floor covering material, improper application of sound deadening material (see 3.5.4, (3.5.4.2 and 3.7.8).	Visual
106	Stepwells	Hinge cover not flush with interior floor (when applicable), binding or interference of hinged cover or fold-out stair assembly (whichever is applicable) damaged, defective material, improper fit (see 3.5.4.1).	Visual and operational
107	Electrical system	Sloppy or loose connections, improper wiring clearances, color code errors, incorrect capacity of circuit breaker box(es),	Visual and functional

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TABLE I. Major defects AQL - 25 defects per 100 units. - Continued

Defect number	Characteristic	Major defects	Method of inspection
		wiring not fastened or clipped, incorrect wire size, faulty materials, faulty solder joints, sloppy installation, inoperative electrical controls, noisy (see 3.5.5 and 3.7.9).	
108	Floodlights installation	Not weatherproof, improper grounding, improper light wattage, sloppy installation, brackets not secure, brackets not located as specified, inoperative (see 3.5.5.4).	Visual and operational
109	Air conditioners, heaters and controls	Air conditioners and heaters not mounted securely, sloppy installation of controls, defective materials, malfunction of controls, vibration and undue noise of heaters and air conditioners (see 3.5.6, 3.5.7 and 3.5.10).	Visual and operational
110	Plumbing system, components and fixtures	Improper assembly or installation, defective material, fixtures and components not mounted securely, malfunction of plumbing fixtures, water leaks, undue noise (see 3.5.8 and 3.7.11).	Visual and operational
111	Laminated surfaces (Formica) desks, counters and cabinets	Delamination, scratches, mars, color fade, unevenness, bulging, rough edges, not level with floor surface, not mounted	Visual and functional

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TABLE I. Major defects AQL - 25 defects per 100 units. - Continued

Defect number	Characteristic	Major defects	Method of inspection
		securely, improper assembly or installation, improper construction, improper counter height, binding and interference of desks and cabinet drawers (see 3.5.9.2 through 3.5.9.4).	
112	Benches with back-rest, stools and chairs	Benches not sturdy or not rigid construction, improper installation, stool and chair legs uneven, defective material, interference with lift-up seat (see 3.5.9.1 through 3.5.9.3 and 3.7.13).	Visual and functional
113	Upholstery and padding	Sloppy installation, inferior material, uncomfortable, nonuniform fit, rips, splits, color fade, stains, color not coordinated with applicable area, material not durable (see 3.5.9.1 through 3.5.9.3 and 3.7.14).	Visual and tactile
114	Grab handles and safety handrails	Improper installation, not mounted securely, not rigid construction, grab handles and handrail surfaces not smooth (burrs, snags, scratches, rough areas) (see 3.5.2.2, 3.5.4.1, and 3.7.15).	Visual and tactile
115	Sound system	Improper installation, loose connections, not installed in proper areas, exposed wiring or insulation, speakers not mounted securely, inconvenient location of volume controls,	Visual and functional

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TABLE I. Major defects AQL - 25 defects per 100 units. - continued

Defect number	Characteristic	Major defects	Method of inspection
		defective materials, malfunction of controls (see 3.5.10 and 3.7.16).	
116	Exhaust fans	Not mounted securely, improper locations, improper fit to interior panels, exposed wiring, exposed insulation, loose connection, improper fit to exterior panels (discharge openings), not weatherproof, sloppy caulking, fans and controls not operative, vibration, rattles, undue noise (see 3.5.9.4(g)).	Visual and functional
117	Curtains	Improper installation, not secure, not even, inferior material, color not appropriate, uneven stitching, tears, frays, color fade, stains (see 3.5.2.4 and 3.7.6).	Visual
118	Sealers and caulking	Sloppy application, improper amounts, improper material, not weatherproof or waterproof.	Visual
119	Fire extinguishers	Location not convenient, mounting not secure.	Visual

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for use at various installations in the continental United States; for trips over streets, highways and gravel roads; and for use as a self-contained mobile blood donor facility and blood component processing laboratory.

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* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Special exterior markings, if required (see 3.1.1.2).
(d) Rustproofing, if required (see 3.1.1.3).
(e) Tropical rustproofing, if required (see 3.1.1.3).
(f) Power plant heaters and fuel warmer, if required (see 3.4.1.6).
(g) If a battery heater is not required (see 3.4.1.6).
(h) Alternator capacity, if other than as specified (see 3.4.2.1).
(i) Automatic transmission, if required (see 3.4.5.2).
(j) Two-speed axle, if required (see 3.4.9.1).
(k) Traction control, if required (see 3.4.9.2).
(1) Disc type wheels, if required (see 3.4.10).
(m) Bias ply tires, if required (see 3.4.10.1).
(n) If a spare tire carrier is not required (see 3.4.10.3).
(o) Spare wheel or rim, if required (see 3.4.10.4).
(P) Spare tire assembly, if required (see 3.4.10.5).
(q) A desiccant type air dryer, if required (see 3.4.11.l(k)).
(r) Tools , if required (see 3.4.16.1).
(s) Air operated horn, if required (see 3.4.22).
(t) Automatic speed control, if required (see 3.4.23).
(u) If exterior floodlights are not required (see 3.5.5.4).
(v) Additional electric heaters, if required (see 3.5.7).
(w) External water supply capability, if required (see 3.5.8).
(x) Water heater, if required (see 3.5.8).
(Y) Fresh water and waste equipment, if required (see 3.5.8).
(z) Additional equipment, if required (see 3.5.9.5).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE J688, Truck Ability Prediction Procedure Tables:

Table 1	Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2 – Altitude Factor. 1.00
Table 3 - Rolling Factor. 1.841
Table 4 - Area Factor. 0.250
Table 5 - Velocity Factor. 333
Table 6 - Altitude Factor. 1.00
Table 7 - Chassis Friction Horsepower. Use applicable power unit GVW and the engine revolutions per minute (rpm) (to nearest 100 revolutions) which is required for a geared speed of 55 mph. For engine speed beyond the range of this table, the following formula shall be used:
FHP = 1 + (0.0000002 rpm) (GVW)
Table 8 - Grade Factor. 0.68
Table 8A - Correction Factor. Not required.
Table 9 - Road Factor. 0.0.

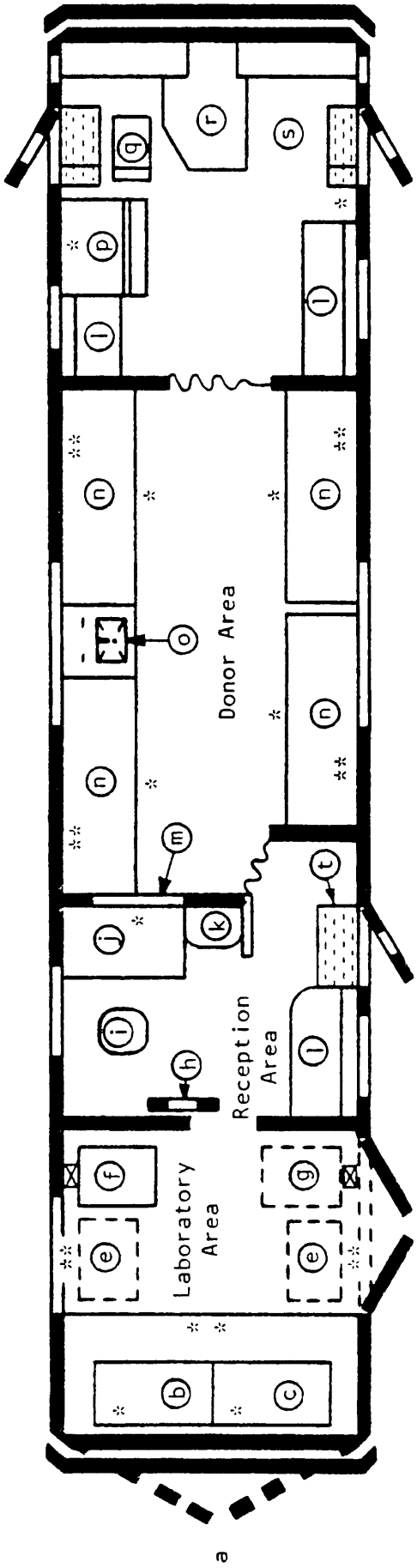
6.4 Identification of changes. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

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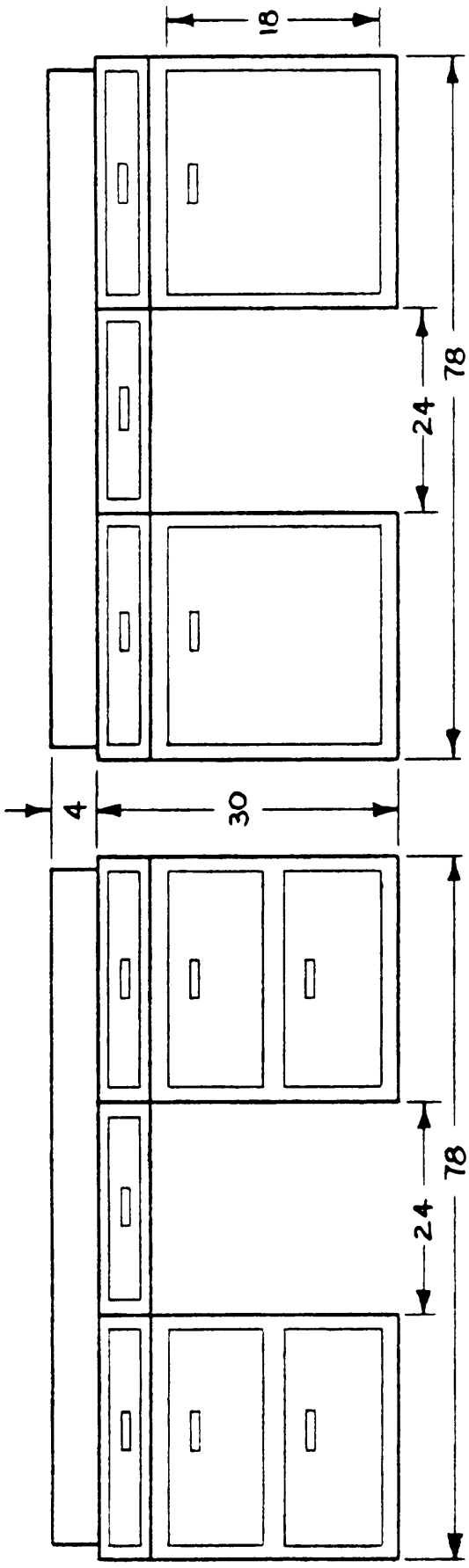


a. Access To 45kW Generator.
b. Blood Bank Refrigerator.
c. Blood Plasma Freezer.
d. Equipment Access Doors.
e. Optional Blood Bank Refrigerator.
f. Refrigerated Centrifuge.
g. Optional Refrigerated Centrifuge.
h. Sliding Door, Opening 24 Inch Min.
i. Stool.
j. Desk.
k. Chair.
l. Padded Bench.
m. 24 x 36 Inch One Way Glass.
n. Donor Bed and Counter.
o. Sink.
p. Refrigerator.
q. Driver's Seat.
r. Padded Seat Over Engine.
s. Forward Tilt Passenger Seat.
t. Pull-out Step Assembly.

* = 120 Volt Duplex Outlet
⊠ = 240 Volt Outlet

FIGURE 1. Interior arrangement.

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NOTES: 1. Style 1 and Style 2 cabinets are 27 inches deep.
2. All dimensions are in inches.

FIGURE 2. Storage cabinet and drawer arrangement.

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8. GENERAL COMMENTS:

9. RECOMMENDED CHANGE(S) TO PUBLICATION

PAGE NO.	PARA-GRAPH	LINE NO.	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASONS

10. ORIGINATOR	11. RANK/RATE OR GRADE AND TITLE	12. DATE

13. WORK CENTER	14. TELEPHONE (AUTOVON/COM'L)

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